

SEAT No. _____

SARDAR PATEL UNIVERSITY

[9/10]

B.Sc.(CA&IT) (Regular)/ M.SC(IT) Integrated (NC)**Sem-II EXAMINATION****MONDAY, 9th APRIL, 2018****10:00 am to 12:00 noon****SUBJECT: MATHEMATICS-II (PS02FIIT02)**

Maximum Marks: 70

Q-1 Write the correct option in the answer book.

[10]

- (1) A square of is called Variance
 (a) Range (b) Mean deviation (c) Standard deviation (d) Quartile deviation
- (2) The number of elements in a set $\{x \in \mathbb{N}: x^2 - 2 = 0\}$ are:
 (a) 1 (b) 2 (c) $\pm\sqrt{2}$ (d) 0
- (3) Quartile deviation is defined as
 (a) $\frac{Q_3 - Q_1}{2}$ (b) $\frac{Q_3 - Q_1}{3}$ (c) $\frac{Q_1 - Q_2}{2}$ (d) $\frac{Q_1 + Q_1}{2}$
- (4) De Morgan's Law:
 (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = (A \cap B)^c$
 (c) $(A \cup B)^c = A \cap B$ (d) None
- (5) $P(7, 3) =$
 (a) 21 (b) 12 (c) 120 (d) 210
- (6) Let $A = \{0, 1\}$, then A closed under:
 (a) multiplication (b) addition (c) division (d) subtraction
- (7) Every group are:
 (a) Abelian group (b) ring (c) semi group (d) none
- (8) 4 pens can be selected from 8 pens in..... ways.
 (a) 8! (b) 1680 (c) 70 (d) $\frac{8!}{4!}$
- (9) The set $\{x \in \mathbb{R}: 2 < x < 79\}$ is:
 (a) finite (b) infinite (c) empty (d) none
- (10) The set $\{\mathbb{N}, +\}$ is:
 (a) group (b) ring (c) monoid (d) semi group

Q:2 Answer the following in short. (Any Ten)

[20]

- (1) Define the terms: Group, Monoid.
- (2) In $\mathbb{Z}_{10}$, find -3 , -8 , 3^{-1} , 5^{-1} .
- (3) If S is a nonempty set with the operation $a*b = b$. Is the operation* associative?
- (4) Define range and quartile deviation.

(PTO)

- (5) Find the number of distinct permutation that can be formed from all the letters of the word: (i) RADAR (ii) NAAC.
- (6) If the quartiles $Q_1 = 62$, $Q_2 = 142$ and $Q_3 = 195$, then find the Bowley's coefficient of skewness.
- (7) If $A = \{1, 2, 3, \dots, 12\}$ and $B = \{4, 6, 7, \dots, 16\}$ then find the symmetrical difference between A and B.
- (8) Find dual of the following:
 (i) $(A \cap B \cup C)^c = (A \cap C)^c \cap (A \cap B)^c$, (ii) $(A \cap \phi) \cup (U \cap A^c) = A$.
- (9) Find the number of ways that a party of 5 persons can arrange themselves: (i) around a circular table and (ii) in a row.
- (10) Define power and find the power set of a set $\{x, y, z\}$.
- (11) How many committees of 3 with a given chair person can be selected from 8 persons?
- (12) Find variance of the observations: 4, 4, 4, 4, 4, 4, 4, 4.

Q-3

- (a) Let n denote a positive integer. Suppose a function L is defined as [05]

$$L(n) = \begin{cases} 0 & \text{if } n=1 \\ L([n/2]) + 1 & \text{if } n > 1 \end{cases} \text{ Find } L(25) \text{ and } L(34).$$

- (b) Find inverse of the function $f(x) = \frac{7x-3}{5x-2}$, $x \neq \frac{2}{5}$. [05]

Q-3

OR

- (c) Prove that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$. [05]

- (d) By using algebra of sets, prove that: (i) $(\phi \cup A) \cap (B \cup A) = A$, [05]
 (ii) $A \cup (A \cap B) = A$.

Q-4

- (a) For a, b rational number, define $a * b = a + b - ab$. Find identity element for $*$ and if $a \neq 1$, then find a^{-1} . [05]

For $a, b \in Q$ (rational numbers), define $a * b = a + b + ab/2$. [05]

- (b)
 (i) Is $(Q, *)$ Semigroup?
 (ii) Is $(Q, *)$ Monoid?
 (iii) Find its inverse if it exist.

Q-4

- (c) Show that the set $\{0, 1, 2, 3, 4, 5, 6\}$ is a group under addition modulo 7. [05]

- (d) For a, b rational number, define $a*b = ab/5$. Is $(Q, *)$ commutative? Show that $(Q, *)$ is Monoid and find its inverse if it exist. [05]

Q-5

OR

- (a) A debating team consists of 3 boys and 3 girls. Find the number of ways they can sit in a row where: (a) there are no restrictions; (b) the boys and girls are each to sit together; (c) just the girls are to sit together. [05]
- (b) A student is to answer eight out of ten questions on an exam. Find the number m of ways that the student can chose eight questions. Also find m if the student must answer at least four out of the first five questions. [05]

Q-5

OR

- (c) There are twelve points $A, B, \dots, L$ in a given plane, no three on the same line (i) how many triangles are determined by the points? [05]
- (ii) how many of these triangles contain the point A as vertex?
- (d) A class consist of seven men and five women, (i) Find the number m of committees of five that can be selected from the class (ii) Find the number m of committees of five if it consists of three men and two women. [05]

- Q.6 Find regression equations (i) x on y (ii) y on x , from the following data. [06]

(a)

X:	10	6	8	16	12	13	11	9
Y:	19	12	16	39	29	32	24	14

- (b) Explain the linear regression and state its properties. [04]

OR

- Q.6 Following table shows the frequency distribution of marks. [06]

(c)

Marks :	0- 10	10-20	20-30	30-40	40-50	50-60
No. of students:	12	18	27	20	17	6

Find the Mean deviation from mean for above frequency distribution.

- (d) Given below is the frequency distribution of the marks obtained by 90 students. Compute the standard deviation. [04]

Marks:	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
No. of students:	5	12	15	20	18	10	6	4

